Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2013-0402-DS-3-RF-101	Issued	2013-09-09

UT Thickness Data Sheet - 3-RF-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 3-RF-101, located within the ATU-100. The inspection was conducted by Kevin Doucette, acting as the lead UT technician, to support the 3-year regenerator overhead and reflux survey.

The primary objective is to assess the current wall thickness of the reflux line against the established t-min of 2.41 mm to determine the remaining life and next inspection interval.

Technical Specifications

Line 3-RF-101 is constructed from A312 TP316 with a nominal thickness of 5.49 mm and an outer diameter of 88.9 mm. The design limits for this circuit are 517 kPa(g) and 104 °C.

The line is categorized under Class 2 for amine, reflux, and treated gas, which dictates a thickness measurement interval of 10 years and an external visual interval of 5 years.

Procedural Compliance

Thickness measurements were performed in accordance with HIS-UT-002 Rev 1 and the site-specific QSCP-NDT-001 Rev 2. The latter procedure incorporates requirements for the CUI program, including insulation removal and inspection-port specifications.

The inspection methodology and equipment calibration were executed as per the guidelines defined in ASTM E797.

Inspection Results and Analysis

The data collected for 3-RF-101 is detailed in the associated report HIS-2013-0402. All readings are compared against the retirement thickness of 2.41 mm.

Corrosion rates are calculated based on the larger of the long-term rate or the short-term rate from the latest pair of readings at least 3 years apart. Any reading falling below the t-min is subject to an NCR.

Thickness readings, measured 2013-09-09

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2010-09-13	6.78	6.49	0.063	0.077	38.1	2016-09-09	
3-RV-101-02	7.62	3.56	2010-09-13	6.59	6.44	0.065	0.075	38.3	2016-09-09	
3-RV-101-03	7.62	3.56	2010-09-13	6.16	5.80	0.098	0.127	17.6	2016-09-09	
3-RV-101-04	7.62	3.56	2010-09-13	6.43	6.13	0.083	0.089	29.0	2016-09-09	
3-RV-101-05	7.62	3.56	2010-09-13	6.63	6.48	0.068	0.065	43.0	2016-09-09	
3-RV-101-06	7.62	3.56	2010-09-13	6.43	6.11	0.085	0.100	25.4	2016-09-09	
3-RV-101-07	7.62	3.56	2010-09-13	6.20	5.73	0.157	0.157	13.8	2016-09-09	
3-RV-101-08	7.62	3.56	2010-09-13	6.71	6.36	0.117	0.117	23.9	2016-09-09	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2010-09-13	4.55	4.49	0.059	0.057	28.6	2016-09-09	
2-RV-102-02	5.54	2.79	2010-09-13	4.60	4.29	0.074	0.064	20.2	2016-09-09	
2-RV-102-03	5.54	2.79	2010-09-13	4.84	4.68	0.046	0.055	34.2	2016-09-09	
2-RV-102-04	5.54	2.79	2010-09-13	4.84	4.81	0.044	0.052	38.9	2016-09-09	
2-RV-102-05	5.54	2.79	2010-09-13	5.04	4.96	0.027	0.027	>50	2016-09-09	
2-RV-102-06	5.54	2.79	2010-09-13	4.82	4.55	0.090	0.090	19.5	2016-09-09	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2006-02-24	5.48	5.39	0.005	0.012	>50	2023-09-09	
3-RF-101-02	5.49	2.41	2006-02-24	5.37	5.34	0.006	0.004	>50	2023-09-09	
3-RF-101-03	5.49	2.41	2006-02-24	5.45	5.45	0.000	0.000	>50	2023-09-09	
3-RF-101-04	5.49	2.41	2006-02-24	5.33	5.33	0.007	0.000	>50	2023-09-09	
3-RF-101-05	5.49	2.41	2006-02-24	5.40	5.26	0.011	0.019	>50	2023-09-09	
3-RF-101-06	5.49	2.41	2006-02-24	5.40	5.25	0.015	0.020	>50	2023-09-09	
3-RF-101-07	5.49	2.41	2006-02-24	5.35	5.43	-0.001	-0.011	>50	2023-09-09	
3-RF-101-09	5.49	2.41	—	—	5.12	—	—	—	2023-09-09	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2006-02-24	3.85	3.71	0.012	0.019	>50	2023-09-09	
2-RF-102-02	3.91	2.16	2006-02-24	3.88	3.65	0.013	0.031	48.8	2023-09-09	
2-RF-102-03	3.91	2.16	2006-02-24	3.80	3.76	0.010	0.005	>50	2023-09-09	
2-RF-102-04	3.91	2.16	2006-02-24	3.81	3.80	0.004	0.001	>50	2023-09-09	
2-RF-102-05	3.91	2.16	—	—	3.69	—	—	—	2023-09-09	
2-RF-102-06	3.91	2.16	—	—	3.71	—	—	—	2023-09-09	
2-RF-102-07	3.91	2.16	2006-02-24	3.85	3.82	0.008	0.004	>50	2023-09-09	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2013-09-09	_____